

SAM-Med: Decoder Fine-tuning of Segment Anything for Polyp Segmentation

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Final Test Dice

0.9374

+0.1189 over zero-shot

Trainable Parameters

4.3%

mask decoder only

Dataset

Kvasir-SEG

800 / 100 / 100 split

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Motivation & Problem Definition

Clinical Need

Early detection

colorectal cancer prevention

Task

Segmentation

pixel-level polyp boundaries

Goal

Adapt SAM

medical-domain fine-tuning

What is a polyp? A small lump in the colon that can become cancer.

The Problem: Doctors can easily miss small or flat polyps during exams.

Why Segmentation? It shows the exact size, shape, and edges.

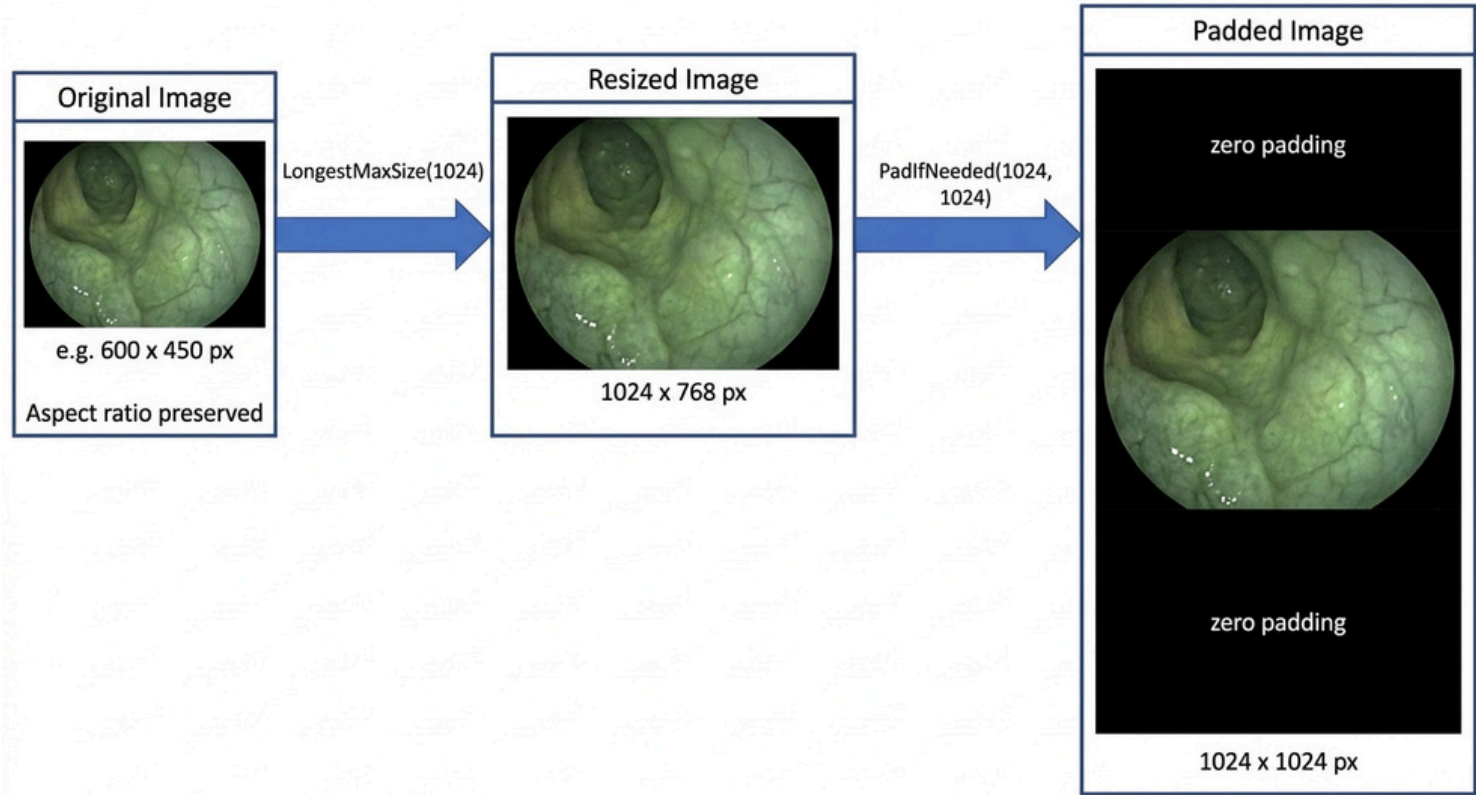
The Challenge: SAM is made for normal photos, not medical images.

Our Solution: Train only a small part of SAM (the decoder) to make it work.

Dataset Overview

Property	Value
Dataset	Kvasir-SEG
Images	1000 colonoscopy frames
Split	800 train / 100 validation / 100 test
Labels	Expert binary polyp masks

Image Preprocessing



Training Augmentations

Prevent overfitting:

- Horizontal and vertical flips
- 90-degree rotations
- Color jitter
- Mask-aligned transforms

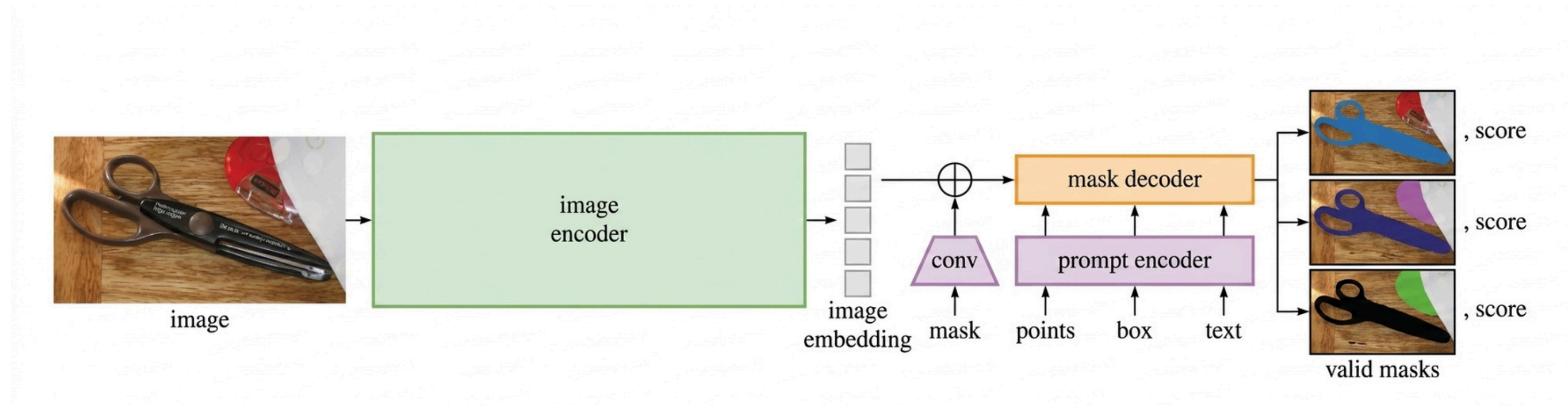
Main Goal: Fit SAM's input size while keeping the real shape of the polyp.

Step 1 (Resize): Scale the longest side to 1024 pixels (keeps the original shape).

Step 2 (Pad): Add black borders (zero padding) to the top and bottom.

Result: A perfect 1024×1024 square without stretching the polyp.

Model Architecture



SAM (Segment Anything Model) Architecture

Image Encoder: Converts raw images into embeddings

Prompt Encoder: Encodes prompt inputs

Mask Decoder: Fuses embeddings to predict masks in real-time

Output: Generates multiple valid masks and confidence scores

Proposed Method

Step	Stage	Description
1	Image	Resize/pad to SAM input
2	Box prompt	BBox from GT mask
3	Frozen encoders	Image + prompt embeddings
4	Mask decoder	Train with BCE + Dice
5	Evaluation	Same bbox protocol for all models

Protocol Detail

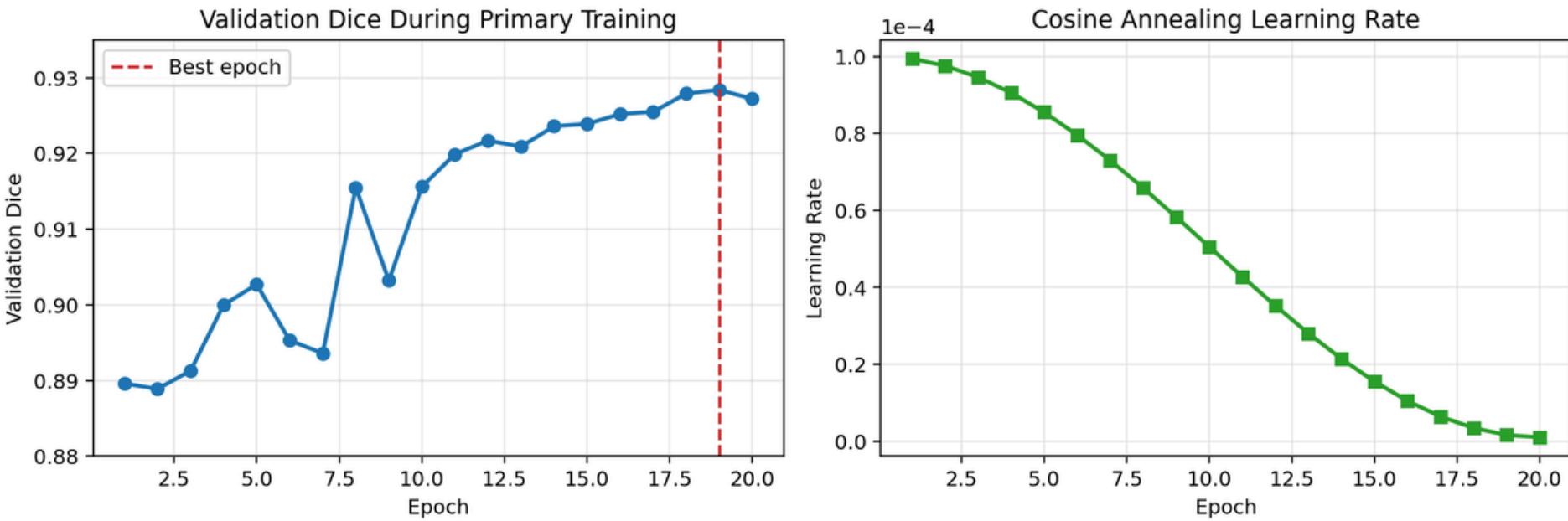
The bbox prompt does not crop the image. It tells SAM where to focus. Zero-shot and fine-tuned models use the same prompt source.

SAM Components

Image encoder: frozen
 Prompt encoder: frozen
 Mask decoder: trainable
 4.06M / 93.7M parameters updated
 Efficient adaptation without full fine-tuning

Training Setup

Hyperparameter	Value
Optimizer	AdamW
Learning rate	1e-4
Weight decay	1e-4
Scheduler	CosineAnnealingLR
Batch size	2
Epochs	20
Loss	0.5 BCE + 0.5 Dice



Best checkpoint: epoch 19, validation Dice = 0.9284

Evaluation Metrics

Dice Coefficient

$$\text{Dice} = 2|P \cap G| / (|P| + |G|)$$

Common in medical segmentation; rewards prediction-ground truth overlap.

Intersection over Union

$$\text{IoU} = |P \cap G| / |P \cup G|$$

Stricter measure; penalizes region and boundary errors more strongly.

100 held-out Kvasir-SEG test images
 Ground-truth bbox prompt for zero-shot and fine-tuned SAM
 Reported metrics are mean Dice and mean IoU

Quantitative Results

Model / Variant	Dice	IoU	Takeaway
Zero-shot SAM	0.8186	0.7595	Not domain-adapted
Fine-tuned SAM	0.9374	0.8925	Final model
Promptless	0.7772	0.6795	Prompt is essential
Morphology	0.9384	0.8938	Tiny gain
CRF	0.8670	0.7860	Hurts performance

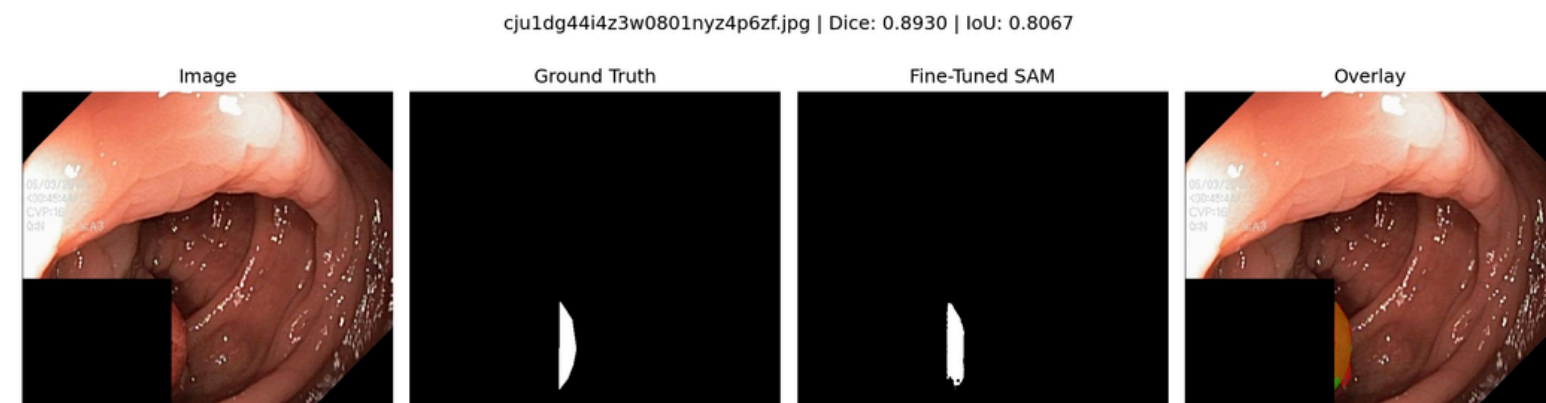
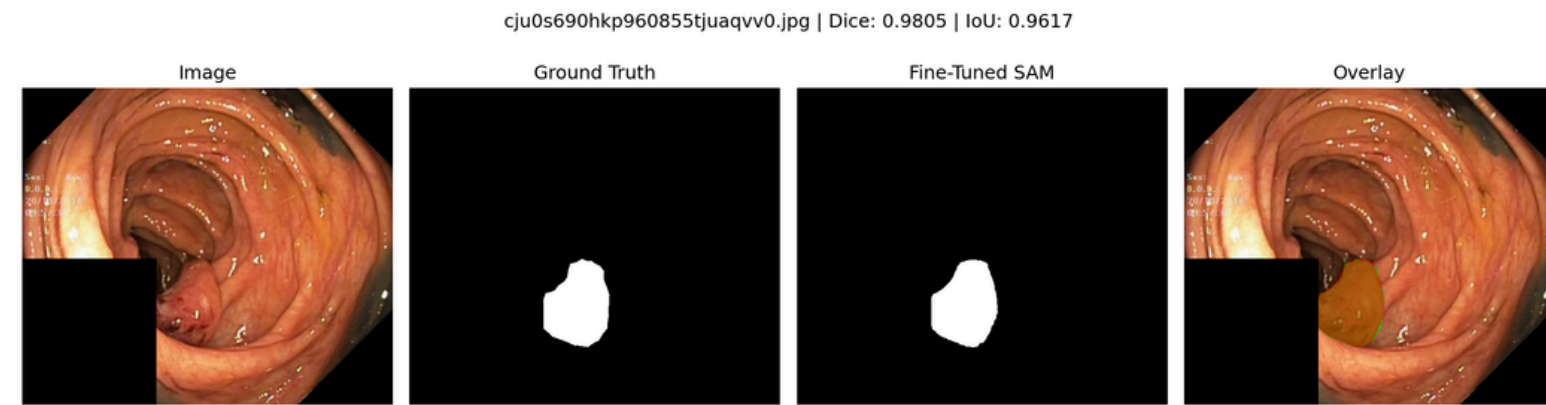
Dice Gain
+0.1189
 fine-tuned vs zero-shot

IoU Gain
+0.1330
 fine-tuned vs zero-shot

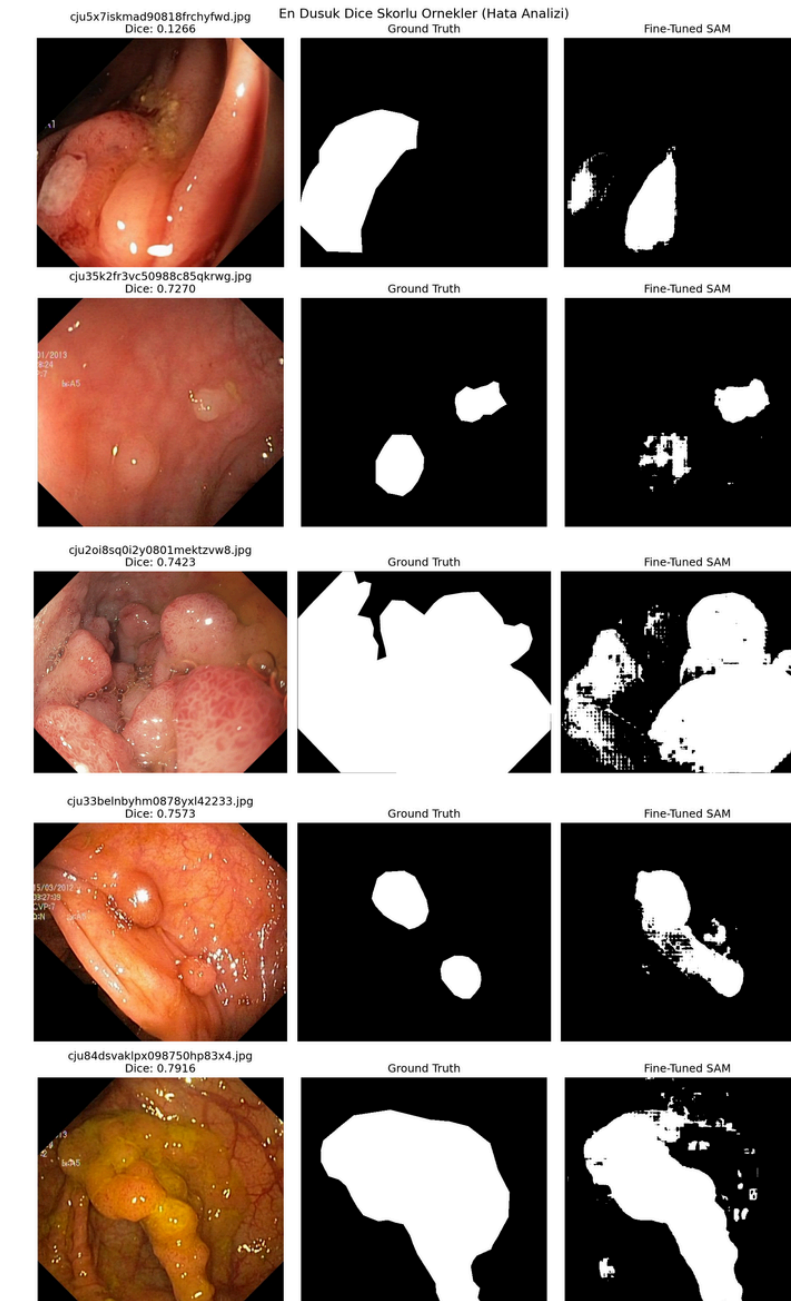
Ablation	Dice	Delta
Baseline	0.9374	-
BCE only	0.9271	-0.0103
Dice only	0.9327	-0.0048
No augmentation	0.9319	-0.0056
High LR	0.9184	-0.0190
Prompt encoder unfrozen	0.9364	-0.0010

Qualitative Results

Representative predictions



Worst-case analysis



Failures mainly involve flat lesions, specular highlights, low contrast, or loose bounding boxes.

Discussion

What Worked

Decoder-only fine-tuning gives a large gain with few trainable parameters.
 BCE + Dice outperforms either loss alone.
 Augmentation improves generalization.
 BBox prompts remain highly informative.

What Did Not Help

High learning rate reduced performance.
 Unfreezing prompt encoder did not improve results.
 Extended training saturated and slightly regressed.
 CRF struggled on homogeneous endoscopy textures.

Limitations & Conclusion

Limitations

Ground-truth bbox prompts are not available in real clinical deployment.
 Flat or sessile polyps remain difficult.
 Specular reflections and low contrast can confuse boundaries.
 Experiments are limited to Kvasir-SEG.

Conclusion

Dice improves from 0.8186 to 0.9374.
 Only 4.3% of SAM parameters are trained.
 Ablations validate the final design.
 Future work: automatic detector + SAM, point prompts, LoRA, cross-dataset validation.

Final message: parameter-efficient SAM adaptation is effective, but deployment may need automatic prompting.

Thank You

Thank you for your attention!

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